



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL TEST

20-12-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	C	B	A,B,C	B,C	A,C,D	A,B,D	A,B,C,D	A,B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6		
	A.	1.28 to 1.29	1.33	1.50	0.79 to 0.80	30.00	9.80		
SECTION-II (ii)	Q.	7	8	9	10				
	A.	11.25	4.00	8.00	3.11 to 3.13				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C,D	A,B,C,D	A,C	C	A,B,C	A,B,C	C	A,B
SECTION-II (i)	Q.	1	2	3	4	5	6		
	A.	4.50	8.49	57.00 to 58.00	0.40	264.00	57.50		
SECTION-II (ii)	Q.	7	8	9	10				
	A.	4.00	8.00	3.00	4.00				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	A,B	A,B	B,C,D	A,B,C	A,C,D	A,B	A,C	A,B,D
SECTION-II (i)	Q.	1	2	3	4	5	6		
	A.	1.00	0.00	2.47 to 2.48	1.78 to 1.79	2.50	3.55 to 3.56		
SECTION-II (ii)	Q.	7	8	9	10				
	A.	3.00	10.00	2.00	1.00				

HINT – SHEET

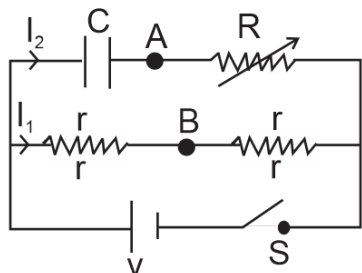
PART-1 : PHYSICS

SECTION-I

1. Ans (C)

$$V_1 = \sum \int_B^1 E_1 dr, V_2 = \int_A^2 E_2 dr \text{ and } E_1 > E_2$$

2. Ans (B)



Consider circuit at any time t after closing the switch :

$$I_1 = \frac{V}{2r} \text{ and } I_2 = \frac{V}{R} e^{-\frac{1}{RC}}$$

$$\Rightarrow V_A - V_B - RI_2 - rI_1$$

$$= R \times \frac{V}{R} e^{-\frac{1}{RC}} - r \times \frac{V}{2r}$$

$$\Rightarrow V_A - V_B = -\frac{V}{2} \left(1 - 2e^{-\frac{1}{RC}} \right)$$

$$\Rightarrow a = -\frac{V}{2} \text{ and } b = \frac{V}{2}$$

$$\text{When } V_A - V_B = 0$$

$$t_0 = RC \ln(2)$$

Hence greater the R , greater the t

5. **Ans (A,C,D)**

$$P_1 = P_2; Z_1 = Z_2$$

$$\therefore \sqrt{(X_L - X_C)^2 + R^2} = \sqrt{X_L^2 + R^2}$$

$$X_C = 2X_L$$

In L-C-R

$$\tan \phi = \frac{X_C - X_L}{R}$$

In L-R circuit

$$\tan(90^\circ - \phi) = \frac{X_L}{R}$$

$$\therefore X_L = R$$

$$P = \frac{E^2}{Z^2} R$$

7. **Ans (A,B,C,D)**

$$\delta = (\mu - 1)A = (1.5 - 1)0.9^\circ = 0.45^\circ = \frac{\pi}{400} \text{ rad}$$

$$\frac{\text{no. of photons}}{\text{time}} = \frac{7.2}{\frac{1240}{2.21} \times 1.6 \times 10^{-19}} \approx 8 \times 10^{18}$$

Momentum incident per sec.

$$= p = \frac{\text{power}}{c} = \frac{7.2}{3 \times 10^8} = 2.4 \times 10^{-8} \text{ N}$$

Magnitude of change in momentum per second along y-axis

$$|\Delta p_y| = p \sin \delta \approx p\delta = 2.4 \times 10^{-8} \times \frac{\pi}{400} = 6\pi \times 10^{-11}$$

for photons $\Delta \vec{p}_y$ is along $(-\hat{j})$

$\Rightarrow$ for prism $\vec{F}$ is along $\hat{j}$

PART-1 : PHYSICS

SECTION-II (i)

1. **Ans (1.28 to 1.29)**

Let voltage drop across L_1 , R, C and L_2 is a, b, c, d then

$$a^2 + b^2 = 152$$

$$b^2 + c^2 = 20^2$$

$$|c - d| = 9$$

2. **Ans (1.33)**

$$(a + b - c)^2 + b^2 = 12^2$$

$$\Rightarrow a^2 + (c - d)^2 - 2a(c - d) + b^2 = 12^2$$

$$15^2 + 9^2 - 2a(c - d) = 12^2$$

$$2a(c - d) = 2 \times 92$$

$$c - d = 9, a = 9, b = 12, c = 16, d = 7$$

PART-1 : PHYSICS

SECTION-II (ii)

8. **Ans (4.00)**

$$E = 4(1 - 2x) = 0 \Rightarrow x = \frac{1}{2}$$

If the particle crosses the position $x = \frac{1}{2}$, it will move to infinity

From work-energy theorem,

$$-q \int_0^{1/2} 4(1 - 2x) dx = 0 - \frac{1}{2}mv_0^2$$

$$\Rightarrow 4q[x - x^2]_0^{1/2} = \frac{1}{2}mv_0^2$$

$$\Rightarrow v_0 = \sqrt{\frac{2q}{m}} = \sqrt{2 \times 8} = 4 \text{ m/s}$$

9. **Ans (8.00)**

$$f = \frac{3v}{4(L + 0.6r)}$$

$$\frac{df}{dt} = \frac{3v}{4} \left(-\frac{1}{(L + 0.6r)^2} \cdot (0.6) \frac{dr}{dt} \right)$$

$$-2 = \frac{3v}{4} \left(0.6 \frac{dr}{dt} \right)$$

$$\frac{8}{3v \times 0.6} = \frac{dr}{dt}$$

$$\frac{dr}{dt} = \frac{1}{72} \text{ m/s}$$

PART-2 : CHEMISTRY

SECTION-I

1. **Ans (A,B,C,D)**

NaCl in dilute solution ionizes to 100% extent.

$$\therefore i \text{ for NaCl} = 1 + (2 - 1)\alpha = 1 + (2 - 1)1 = 2$$

Total moles of Na^+ and Cl^- ions = $2 \times 0.5 = 1$

$$\frac{P^\circ - P_o}{P^\circ} = \frac{n_2}{n_1 + n_2}$$

Relative lowering of vapour pressure of final

$$\text{solution} = \frac{760 - 750}{760} = \frac{10}{760} = \frac{1}{76}$$

$$\therefore \frac{1}{76} = \frac{n_2}{n_1 + n_2} \Rightarrow \frac{n_1}{n_2} = \frac{75}{1} \Rightarrow n_1 = 75 \text{ (as } n_2 = 1)$$

(A) Water separated as ice = $150 - 75 = 75$

moles

= $75 \times 18 = 1.35 \text{ kg}$

$$(B) \Delta T, \text{ of final solution} = k_f \times \frac{1 \times 1000}{75 \times 18}$$

$$2 \times \frac{1000}{75 \times 18} = \frac{40}{27}$$

$$\text{Freezing point} = \left(273 - \frac{40}{27} \right) K$$

(C) Freezing point of original solution =

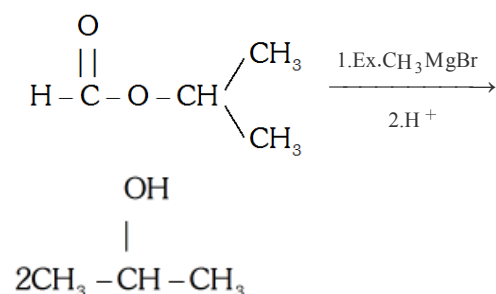
$$\left[273 - \frac{2 \times 1000}{150 \times 18} \right]$$

$$= \left(273 - \frac{20}{27} \right) K$$

(D) Relative lowering of vapour pressure of

$$\text{final solution} = \frac{760 - 750}{760} = \frac{10}{760} = \frac{1}{76}$$

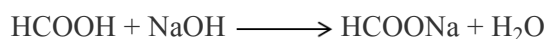
8. **Ans (A,B)**



PART-2 : CHEMISTRY

SECTION-II (i)

1. **Ans (4.50)**



$$\begin{array}{ccc} 30 & 20 & 0 \\ 10 & 0 & 20 \end{array}$$

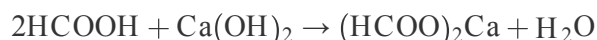
$$\text{pH} = \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

$$= 4.2 + \log \frac{20}{10}$$

$$= 4.2 + 0.3010$$

$$= 4.5$$

2. **Ans (8.49)**



$$\begin{array}{ccc} 10 & 5 & 0 \\ 0 & 0 & 5 \end{array}$$

Salt hydrolysis of (SB and WA)

initially milli moles of $\text{HCOO}^- = 20$

Total milli moles $\text{HCOO}^- = 20 + 10 = 30$

3. **Ans (57.00 to 58.00)**

If mole fraction of liquid A in vapour is = x

Mole fraction of liquid A in solution is = 1 - x

Mole fraction of liquid B in vapour is 1 - x

Mole fraction of liquid B in solution is = x

$$Y_A = \frac{P_A}{P_T} = \frac{P_A^0 X_A}{P_T}$$

$$Y_B = \frac{P_B}{P_T} = \frac{P_B^0 X_B}{P_T}$$

$$x = Y_A = \frac{102(1-x)}{P} (1-x) = Y_B = \frac{32x}{P}$$

$$\text{Dividing } \frac{x}{(1-x)} = \frac{102(1-x)}{32x}$$

$$\frac{x^2}{(1-x)^2} = \frac{102}{32}$$

$$x = 0.641$$

$$Y_B = 0.641 \quad Y_A = 0.359$$

$$X_B = 0.359 \quad X_A = 0.641$$

$$P = P_B^0 X_B + P_T^0 X_T = 102 \times 0.359 + 32 \times 0.64 = 57.13$$

4. **Ans (0.40)**

Solvent A moles = a

Solvent B moles = b

Solute moles = c

At

$$20^\circ\text{C} \frac{a}{a+b+c} \times 15 + \frac{b}{a+b+c} \times 85 = 40 \quad \dots(1)$$

At

$$30^\circ\text{C} \frac{a}{a+b+c} \times 34 + \frac{b}{a+b+c} \times 136 = 68 \quad \dots(2)$$

On simplifying we get a = b then using eq. (1)

we get a + b + c = 2.5 b = 2.5 a

$$\frac{a}{a+b+c} = 0.4 \rightarrow \frac{b}{a+b+c} = 0.4$$

$$C = 0.2$$

$$\text{Solvent } X_A = 0.4$$

$$\text{Solvent } X_B = 0.4$$

$$\text{Solute} = 0.2$$

5. **Ans (264.00)**

Acetone obeys Henry law and chloroform obeys Raoult's law.

$$P_T = 0.05 \times 150 + 0.88 \times 270 = 264 \text{ Torr}$$

6. **Ans (57.50)**

$$50 = P_{\text{Methanol}}^0 \times 0.04 + 0.96 \times 20$$

$$P_{\text{Methanol}}^0 = 770 \text{ Torr}$$

$$P_T = 770 \times 0.05 + 0.95 \times 20 = 57.50 \text{ Torr}$$

PART-2 : CHEMISTRY

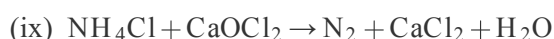
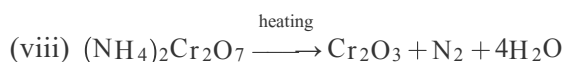
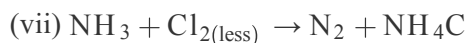
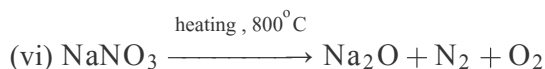
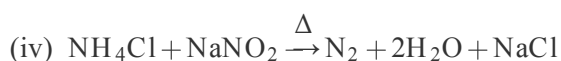
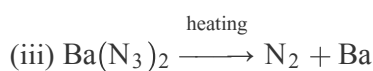
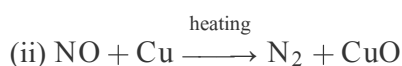
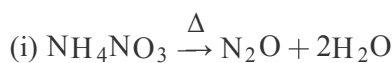
SECTION-II (ii)

7. **Ans (4.00)**

Aniline on reaction with CHCl_3/KOH , $\text{CS}_2/\text{HgCl}_2$, $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$,

and $\text{C}_6\text{H}_5\text{N}_2\text{Cl}^+$ gives phenyl isocyanide, mustard oil reaction, sulfonamide and coupling reaction respectively.

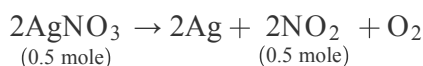
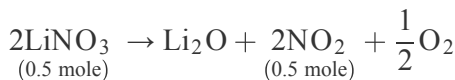
8. **Ans (8.00)**



9. **Ans (3.00)**

6 compounds has 0.5 mole each

NaNO_3 and NH_4NO_3 do not give NO_2



Total number of moles of NO_2 evolved = $0.5 + 1 + 1 + 0.5 = 3$

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A,B)**

By A.M.G.M.

$$\frac{xy + xz + xw + yz + yw + zw}{6} \geq \sqrt[6]{x^3y^3z^3w^3} = \sqrt[6]{xyzw} \quad \dots(1)$$

$$\Rightarrow \sum xy \geq 6\sqrt[6]{xyzw}$$

$$\therefore xyzw = 27 + \sum xy$$

$$\Rightarrow xyzw \geq 27 + 6\sqrt[6]{xyzw} \quad \dots(2)$$

Again by A.M. - G.M.

$$\frac{x + y + z + w}{4} \geq \sqrt[4]{xyzw} \Rightarrow \sqrt[4]{xyzw} \leq 3 \Rightarrow \sqrt[6]{xyzw} \leq 9 \quad \dots(3)$$

from (2) and (3)

$$9 \leq \sqrt[6]{xyzw} \leq 9 \Rightarrow \sqrt[6]{xyzw} = 9$$

(i.e. A.M. = G.M.)

$$\Rightarrow x = y = z = w = 3$$

2. **Ans (A,B)**

$$y = -2 \text{ \& } y = mx \text{ intersects at } \left(-\frac{2}{m}, -2\right)$$

$x = a + m$ also passes through this point

$$\Rightarrow -\frac{2}{m}a + m$$

$$\Rightarrow a = -m - \frac{2}{m} \quad (m < 0)$$

$$\Rightarrow a = \left(\sqrt{-m} - \sqrt{\frac{2}{-m}}\right)^2 + 2\sqrt{2}$$

$$\Rightarrow a \geq 2\sqrt{2}$$

3. Ans (B,C,D)

(a) Not necessary, as $f(x)$ may be discontinuous also

(b) If $x \in (0, \frac{\pi}{2})$ $\sin x < x$ & $\tan x > x$

$$\therefore \frac{\sin x}{x} < 1 < \frac{\tan x}{x}$$

$$(c) \left| \int_a^b (f(x) + g(x)) dx \right|$$

$$\leq \int_a^b (|f(x) + g(x)|) dx$$

$$(d) I = \int_0^{\log 2} [2e^{-x}] dx + \int_{\log 2}^{\infty} 0 dx$$

$$= \int_0^{\log 2} [2e^{-x}] dx = \int_0^{\log 2} [2e^{-(\log 2 - x)}] dx$$

$$= \int_0^{\log 2} [2e^{-(\log 2)} \cdot e^x] dx = \int_0^{\log 2} [e^x] dx$$

4. Ans (A,B,C)

$$(B) P(n=2) = {}^6C_1 \cdot \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{6}$$

$$(C) P(n=4) = 1 - \left({}^6C_4 \left(\frac{1}{6} \right)^4 \right) = \frac{13}{18}$$

$$(D) P(n=6) = 1 - \left({}^6C_6 \cdot \left(\frac{1}{6} \right)^6 \right) = \frac{319}{324}$$

$P(n \geq 7) = 1$ (Numbers will be repeated)

5. Ans (A,C,D)

$$|A| = (1 - \omega^2) - \omega^3(\omega^3 - \omega^3) + \omega^2(\omega^4 - \omega^2)$$

$$\therefore |A| = 1 - \omega^2 + \omega^3 - \omega = 2 - (\omega + \omega^2) = 3$$

$\therefore A$ is non singular

6. Ans (A,B)

$$|3| = |\sin^2 \theta_1 z^4 + \cos^2 \theta_2 z^3 + \sin^2 \theta_3 z^2 + \cos^2 \theta_4 z + \sin^2 \theta_5|$$

$$|3| \leq |\sin^2 \theta_1| |z^4| + |\cos^2 \theta_2| |z^3| + |\sin^2 \theta_3|$$

$$|z^2| + |\cos^2 \theta_4| |z| + |\sin^2 \theta_5|$$

$$3 \leq |z|^4 + |z|^3 + |z|^2 + |z| + 1$$

$$3 < 1 + |z| + |z|^2 + \dots \infty$$

$$3 < \frac{1}{1 - |z|}$$

$$\frac{2}{3} < |z| < 1$$

7. Ans (A,C)

$$N = 10a + b \text{ when } b = na \text{ and } 10a = mb$$

$$\Rightarrow mn = 10$$

$$\therefore (i) m = 10 \text{ and } n = 1 \Rightarrow a = b$$

$$\Rightarrow 9 \text{ numbers}$$

$$(ii) m = 5 \text{ and } n = 2 \Rightarrow b = 2a$$

$$\Rightarrow 4 \text{ numbers}$$

$$(iii) m = 2 \text{ and } n = 5 \Rightarrow b = 5a$$

$$\Rightarrow 1 \text{ number}$$

8. Ans (A,B,D)

$$f(x) = \begin{cases} \frac{1}{x+1} & 0 \leq x < 1 \\ \frac{2}{x} & 1 \leq x < 2 \\ \frac{3}{x-1} & 2 \leq x < \frac{5}{2} \end{cases}$$

clearly $f(x)$ is discontinuous and bijective function

$$\lim_{x \rightarrow 1^-} f(x) = \frac{1}{2}$$

$$\lim_{x \rightarrow 1^+} f(x) = 2$$

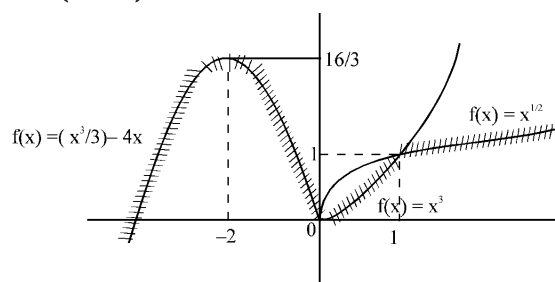
$$\min \left(\lim_{x \rightarrow 1^-} f(x), \lim_{x \rightarrow 1^+} f(x) \right) = \frac{1}{2} \neq f(1)$$

$$\max(1, 2) = 2 = f(1)$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (1.00)



COMMENTS :

function is inc. in $(-\infty, -2) \cup (0, \infty)$

$x = -2 \rightarrow$ local maxima

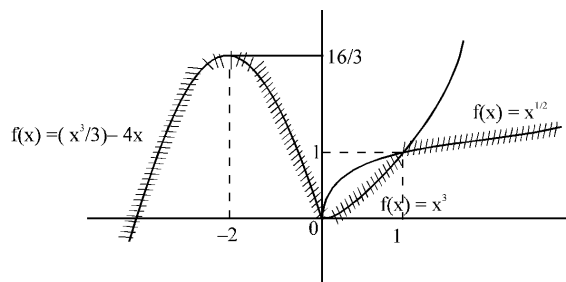
$x = 0 \rightarrow$ local minima

Derivable $\forall x \in \mathbb{R} - \{0, 1\}$ -

$$\left[\begin{aligned} f'(0^+) &= 0, f'(0^-) = -4 \\ f'(1^+) &= 1/2, f'(1^-) = 3 \end{aligned} \right.$$

Continuous $\forall x \in \mathbb{R}$

2. Ans (0.00)



COMMENTS :

function is inc. in $(-\infty, -2) \cup (2, \infty)$

$x = -2 \rightarrow$ local maxima

$x = 2 \rightarrow$ local minima

Derivable $\forall x \in \mathbb{R} - \{0, 2\}$

$$\begin{cases} f'(0^+) = 0, f'(0^-) = -4 \\ f'(2^+) = 1/2, f'(2^-) = 3 \end{cases}$$

Continuous $\forall x \in \mathbb{R}$

3. Ans (2.47 to 2.48)

Since $f(x)$ and $g(x)$ are inverse of each other hence required distance will be twice the distance $f(x)$ and $y = x$

$$\frac{dy}{dx} = 2x = 1$$

$$\Rightarrow x = \frac{1}{2}, y = \frac{9}{4}$$

$$d = 2 \left(\frac{\frac{9}{4} - \frac{1}{2}}{\sqrt{2}} \right) = \frac{7}{2\sqrt{2}}$$

Point on $f(x)$ is $\left(\frac{1}{2}, \frac{9}{4}\right)$

4. Ans (1.78 to 1.79)

similarly for minimum distance $f(x)$ and $y = 2x - 3$

$$\frac{dy}{dx} = 2x = 2 \Rightarrow x = 1$$

point on $f(x)$ is $(1, 3)$

$$\text{required distance } d = \frac{|3 - 2 + 3|}{\sqrt{5}} = \frac{4}{\sqrt{5}}$$

5. Ans (2.50)

$$\text{For coplanar } \begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ a & a^2 & 6 \end{vmatrix} = 0$$

$$\Rightarrow a = 2, -\frac{9}{2}$$

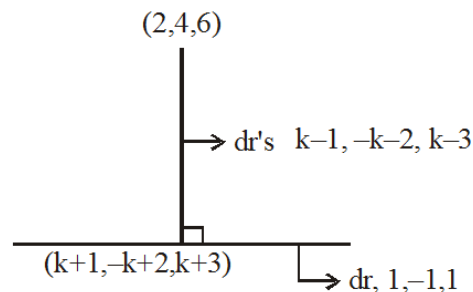
$\therefore$ sum of all possible values of 'a' = $-\frac{5}{2}$

6. Ans (3.55 to 3.56)

$$a \in I \Rightarrow a = 2$$

equation of line

$$L = \frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-3}{1}$$



$$(k-1) + (k+2) + (k-3) = 0$$

$$\Rightarrow k = \frac{2}{3}$$

$\therefore$ perpendicular distance

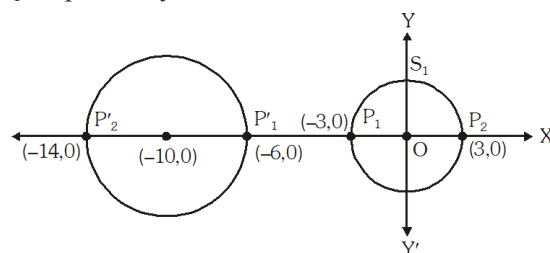
$$= \sqrt{(k-1)^2 + (-k-2)^2 + (k-3)^2} = \sqrt{114}/3$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (3.00)

Clearly, the distance will be minimum between the two ants, when they are at positions P_1 and P'_1 respectively. Hence minimum distance = 3



Also the distance between the two ants will be maximum, when they are at the positions P_2 and P'_2 respectively. Hence minimum distance = 3.

8. Ans (10.00)

$$f(e) = \frac{e}{\sqrt{e^2 - 1}}$$

$f(f(\dots f(e))) = e$, n is even

$$= \frac{e}{\sqrt{e^2 - 1}}, n \text{ is odd}$$

$$\therefore \int_2^6 \frac{e}{\sqrt{e^2 - 1}} de$$

$$\text{Let } e^2 - 1 = t^2$$

$$2ede = 2tdt$$

$$ede = tdt$$

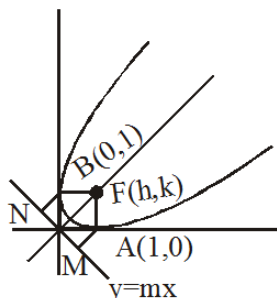
$$\int_{\sqrt{3}}^{\sqrt{35}} dt = \sqrt{35} - \sqrt{3} = k$$

$$\therefore 76k^2 - k^4 = 1024 = 2^p$$

$$p = 10$$

9. Ans (2.00)

Since the x -axis and y -axis are two perpendicular tangents to the parabola and both meet at the origin, the directrix passes through the origin.



Let $y = mx$ be the directrix and (h, k) be the focus.

$$FA = AM$$

$$\Rightarrow \sqrt{(h-1)^2 + k^2} = \left| \frac{m}{\sqrt{1+m^2}} \right| \dots(1)$$

$$\text{And } FB = BN$$

$$\Rightarrow \sqrt{h^2 + (k-1)^2} = \left| \frac{1}{\sqrt{1+m^2}} \right| \dots(2)$$

From equations (1) and (2), we get

$$(h-1)^2 + h^2 + k^2 + (k-1)^2 = 1 \dots(3)$$

and A, B, F are collinear

$$\therefore h + k = 1 \dots(4)$$

By Equation (3) and (4)

focus $F(1/2, 1/2)$

$$\therefore L^2 = 2$$

10. Ans (1.00)

$$e^x(1 + ye^x)(dy + ydx) = 2x dx$$

$$\Rightarrow (1 + ye^x)(e^x dy + ye^x dx) = 2x dx$$

$$\Rightarrow \frac{(1 + ye^x)^2}{2} = x^2 + c$$

$$\Rightarrow ye^x + 1 = \sqrt{2x^2 + 4}$$

$$\therefore f(-1) = (\sqrt{6} - 1)e$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Main)

FULL TEST

20-12-2022

ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	C	D	A	A	A	A	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	C	C	D	A	A	A	B	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	2	625	7	3	4	160	5	28	40	3

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	B	A	A	C	B	D	C	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	C	D	A	C	A	D	D	D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1	3	801	2	7	28	1	2	5	1

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	B	D	D	C	B	D	B	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	B	D	C	C	A	A	A	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1	1	6	2	5	288	8	7	60	5

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (B)**

Theory.

2. **Ans (C)**Speed of light in medium = V_{medium}

$$= \frac{1}{\sqrt{(\epsilon_0 \epsilon_r)(\mu_0 \mu_r)}} = \frac{1}{\sqrt{(\mu_0 \epsilon_0)}} \times \frac{1}{\sqrt{\mu_r \epsilon_r}}$$

$$= \frac{c}{\sqrt{\mu_r \epsilon_r}} \left(c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \right)$$

$$= \frac{c}{\sqrt{2.5 \times 1.6}} = \frac{c}{2}$$

Also, Refractive index of medium

$$= \frac{\text{Speed of light in V vacuum}}{\text{Speed of light in medium}} = \frac{c}{(c/2)} = 2$$

Using Snell's law :

$$\mu_{\text{medium}} \times \sin C = \mu_{\text{vacuum}} \times \sin 90^\circ$$

$$\Rightarrow \sin C = \frac{1 \times 1}{2} = \frac{1}{2} \Rightarrow C = 30^\circ$$

3. **Ans (C)**Let's say $I_0 = KMa^2$

[K = Constant, M = Mass of complete hexagon]

then M.O.I. of removed section of side length

 $a/2$ is

$$I = K \left[m \left(\frac{a}{2} \right)^2 \right] \quad m = \text{mass of small hexagon}$$

$$I_0 = K \left[\sigma \cdot 6 \frac{\sqrt{3}}{4} a^2 \right] a^2 \quad \sigma = \text{Mass p-u area}$$

$$I = K \left[\sigma \cdot 6 \frac{\sqrt{3}}{4} \left(\frac{a}{2} \right)^2 \right] \left(\frac{a}{2} \right)^2$$

$$\therefore \frac{I}{I_0} = \frac{1}{16} \Rightarrow I = \frac{I_0}{16}$$

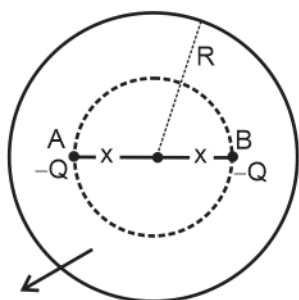
$$\therefore \text{M.O.I. of remaining plate} = I_0 - \frac{I_0}{16} = \frac{15I_0}{16}$$

4. Ans (D)

Let $\rho(r) = Kr$ ($K = \text{constant}$)

$$\text{Total charge} = 4Q = \int_0^R (4\pi x^2 dx)(Kx)$$

$$4Q = 4\pi K \left[\frac{x^4}{4} \right]_0^R \Rightarrow K = \frac{4Q}{\pi R^4}$$



Total charge = $4Q$

Since, both charges do not experience any net force each then

$F_{\text{Due to charge distribution}} = F_{\text{Due to charge } -Q}$

$$EQ \leftarrow \xrightarrow{(B)} \frac{Q^2}{(4\pi\epsilon_0)4x^2} \quad \dots(1)$$

Here, $E = E\text{-field due to charge distribution at location of charge B.}$

To calculate E , using Gauss law

$$\Rightarrow E \times 4\pi x^2 = \frac{1}{\epsilon_0} \int_0^x (Kr)(4\pi r^2 dr)$$

$$= \frac{1}{\epsilon_0} \cdot 4\pi \left(\frac{4Q}{\pi R^4} \right) \left[\frac{r^4}{4} \right]_0^x$$

$$\Rightarrow E = \frac{Qx^2}{\pi\epsilon_0 R^4}$$

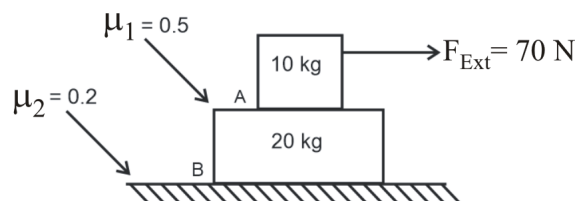
Now, balancing forces on charge B :

$$EQ = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{(2x)^2} \quad (\text{from 1})$$

$$\Rightarrow \left(\frac{Qx^2}{\pi\epsilon_0 R^4} \right) Q = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{(2x)^2}$$

$$\Rightarrow x = R/2$$

5. Ans (A)



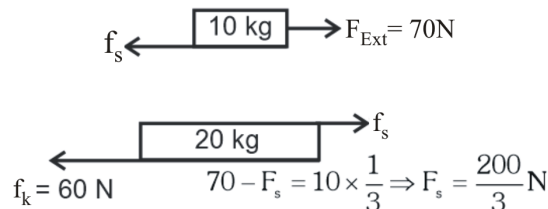
$$(f_{\text{lim}}) = 0.5 \times 100 \text{ N} = 50 \text{ N} \quad (\text{Between A and B})$$

$$(f_{\text{lim}}) = 0.2 \times 300 = 60 \text{ N} \quad (\text{Between B and floor})$$

Assumption - 1 : They both move together as a system.

$$a_{\text{system}} = \frac{70 - 60}{10 + 20} \text{ m/s}^2 = \frac{1}{3} \text{ m/s}^2$$

(Both A and B)



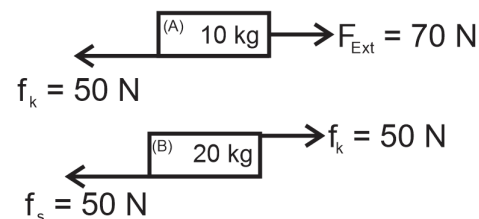
Here, $f_s > f_{\text{lim}}$ or $f_s > 50 \text{ N}$

$\therefore$ Our assumption is wrong.

so, slipping exists between A and B.

$$\text{So, acceleration of A} = \frac{70 - 50}{10} = 2 \text{ m/s}^2$$

towards right and acceleration of B = 0 as static friction by floor on block B will not let it slip.



$$\therefore a_A = 2 \text{ m/s}^2 \text{ towards right and } a_B = 0$$

$$\therefore |\vec{a}_A - \vec{a}_B| = 2 \text{ m/s}^2 \quad \text{Option A}$$

6. Ans (A)

$$P_A + \rho g H + \frac{1}{2} \rho v_A^2 = P_B + \rho g \times 0 + \frac{1}{2} \rho v_B^2$$

$$\Rightarrow \rho g H + \frac{1}{2} \rho v_A^2 = (P_B - P_A) + \frac{1}{2} \rho (4v_A)^2$$

$$\Rightarrow \rho g H + \frac{1}{2} \rho v_A^2 = \rho g (H_2 - H_1) + \frac{1}{2} \rho (16v_A^2)$$

$$\Rightarrow \rho g H - \rho g \frac{H}{16} = \frac{\rho}{2} (16v_A^2 - v_A^2)$$

$$\Rightarrow v_A = \sqrt{\frac{gH}{8}} = \sqrt{\frac{10 \times 0.8}{8}} = 1 \text{ m/sec}$$

$$\therefore \text{Volume flow rate} = \text{Area of cross-section} \times v_A$$

$$v_A = 600 \text{ cm}^2 \times 100 \text{ cm/sec}$$

$$= 60000 \text{ cm}^3/\text{sec}$$

7. Ans (A)

$$\omega_2 - \omega_1 = 2\pi (f_2 - f_1) = \frac{R}{L} \quad (f_1, f_2 \text{ are half power frequencies})$$

$$316 - 296 = \frac{15\Omega}{L} \Rightarrow L = \frac{15}{20} \text{ Henry} = 750 \text{ mH}$$

8. Ans (A)

Energy of wave $\propto$ Intensity

$$\text{Intensity} = \frac{1}{2} \epsilon v E_0^2$$

$\epsilon = \epsilon_0 \epsilon_r$ = permittivity of medium

v = Velocity of EM-wave in medium =

$$\frac{c}{n} = \frac{\text{Speed of light in vacuum}}{R - \text{index of medium}}$$

E_0 = Amplitude of Electric field

$$\text{Energy Transmitted} = \frac{63}{100} \text{ Energy Incident}$$

$$\Rightarrow I\text{-Transmitted} = \frac{63}{100} I_{\text{Incident}}$$

$$\Rightarrow \frac{1}{2} \epsilon_0 \epsilon_r \left(\frac{c}{n} \right) E_0^2 = \frac{63}{100} \times \frac{1}{2} \epsilon_0 c (72 \text{ V/m})^2$$

$$\Rightarrow E_0 = 72 \sqrt{\frac{63}{100} \times \frac{n}{\epsilon_r}} \dots (1)$$

Also, $n = R\text{-Index of medium} = \sqrt{\mu_r \epsilon_r}$

$$= \sqrt{\epsilon_r} \quad (\because \mu_r = 1 \text{ for non-magnetic sapphire})$$

$$\Rightarrow \epsilon_r = n^2$$

$$\therefore E_0 = 72 \sqrt{\frac{63}{100} \times \frac{n}{\epsilon_r}} = 72 \sqrt{\frac{63}{100} \times \frac{n}{n^2}} =$$

$$72 \sqrt{\frac{63}{100} \times \frac{1}{\left(\frac{7}{4}\right)}}$$

$$\therefore E_0 = 72 \sqrt{\frac{36}{100}} = 72 \times \frac{6}{10} = 43.2 \text{ V/m}$$

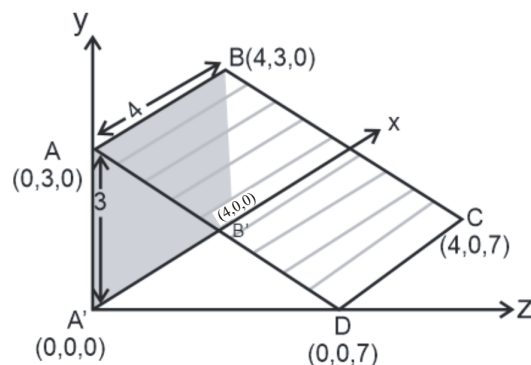
9. Ans (B)

ABB'A' is projection of ABCD on x-y plane.

As the beam is parallel to z-axis, projection ABB'A' is perpendicular to beam.

$\therefore$ Total energy crossing area ABCD in 5 sec.

$$= (\text{Area of ABB'A'}) \times \text{Intensity} \times \text{time}$$



$$= (4 \times 3 \text{ m}^2) \times (200 \text{ Wm}^{-2}) (5 \text{ sec})$$

$$= 12000 \text{ Joules}$$

Energy of single photon

$$= \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{500 \times 10^{-9}} \text{ Joules}$$

$$= 3.98 \times 10^{-19} \text{ J}$$

$\therefore$ number of photons crossing

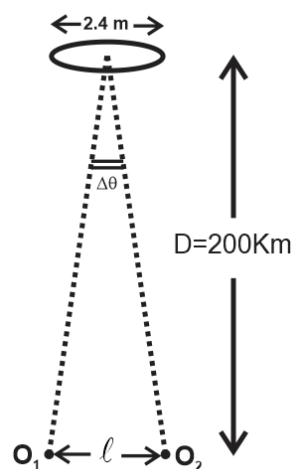
$$= \frac{12000}{3.98 \times 10^{-19}} = 3.01 \times 10^{22}$$

10. Ans (A)

$$\text{For distinct images, } \Delta\theta = \frac{\ell}{D} \geq \frac{1.22\lambda}{a}$$

(a = Aperture diameter)

$$\frac{\ell}{200 \times 10^3 \text{ m}} \geq \frac{1.22 \times 550 \times 10^{-9} \text{ m}}{2.4 \text{ m}}$$



$$\ell \geq 5.59 \times 10^{-2} \text{ m}$$

$$\ell \geq 5.59 \text{ cms} \simeq 5.6 \text{ cms}$$

11. Ans (A)

$$g' = g \left(1 - \frac{2h}{R} \right) \quad \text{For } h \ll R$$

$$\frac{\Delta g}{g} = \frac{g' - g}{g} = \frac{-2h}{R} = -\frac{2 \times 16 \text{ km}}{6400 \text{ km}}$$

$$\therefore \frac{\Delta g}{g} \times 100 \% = -0.5 \%$$

12. Ans (D)

Theory based

13. Ans (C)

$$\rho = \frac{M}{V} \Rightarrow \text{Volume occupied} = \frac{0.25 \text{ kg}}{6 \text{ kg/m}^3}$$

$$= \frac{1}{24} \text{ m}^3$$

$$\text{Total internal Energy} = \frac{f}{2} nRT = \frac{f}{2} PV = \frac{5}{2} \times$$

$$(4 \times 10^4 \text{ N/m}^2) \times \frac{1}{24} \text{ m}^3 = \frac{25}{6} \text{ kJ}$$

14. Ans (C)

Rod AB is undergoing pure rotational motion about axis of platform.

$$\therefore \text{Rotational K.E. of rod} = \frac{1}{2} I_{\text{Axis}} \omega^2$$

$$= \frac{1}{2} \left[\frac{ML^2}{12} + MD^2 \right] \omega^2$$

$$= \frac{1}{2} \left[5 \times \frac{(2\sqrt{3})^2}{12} + 5 \times 3^2 \right] (4 \text{ rad/sec})^2$$

$$= 400 \text{ J}$$

15. Ans (D)

$$\omega_A = \frac{v}{r_A} = \frac{5}{7} \text{ rad/sec},$$

$$\omega_B = \frac{v}{r_B} = \frac{5}{14} \text{ rad/sec}$$

$$\frac{3\pi}{2} = (\omega_A - \omega_B) T$$

$$\Rightarrow \frac{3}{2} \times \frac{22}{7} = \left(\frac{5}{7} - \frac{5}{14} \right) T \Rightarrow T = \frac{66}{5} \text{ sec}$$

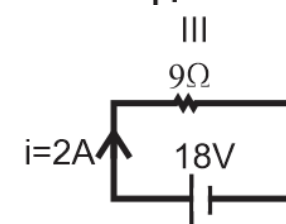
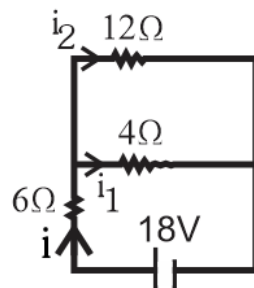
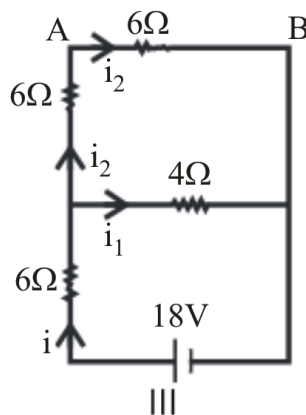
16. Ans (A)

In steady state condition, capacitor will act as open circuit. Potential difference across capacitor = $V_A - V_B$.

$$i_1 + i_2 = i = 2 \text{ A}$$

$$i_1 = \frac{3}{2} \text{ A and } i_2 = \frac{1}{2} \text{ A}$$

$$\therefore V_A - V_B = 6i_2 = 6 \times \frac{1}{2} = 3 \text{ Volts}$$



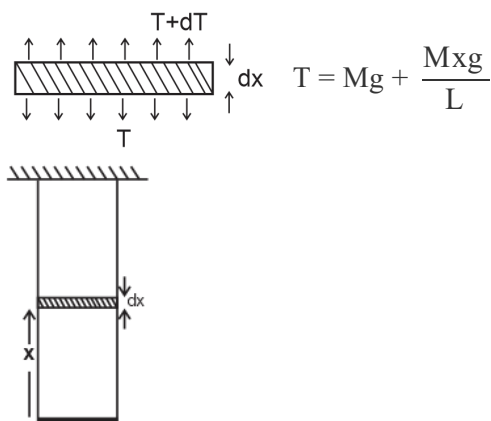
$$\therefore \text{Charge on capacitor} = 5 \mu\text{F} \times (V_A - V_B)$$

$$= 5 \mu\text{F} \times 3 \text{ Volts}$$

$$= 15 \mu\text{C}$$

17. **Ans (A)**

Consider a segment of length 'dx' from lower end of rod.



Extension in this small segment of length 'dx' is

$$d(\text{Extension}) = \frac{T(dx)}{AY} = \frac{1}{AY} \left(Mg + \frac{Mxg}{L} \right) dx$$

∴ Total extension in entire rod = $\int d(\text{Extension})$

$$= \frac{Mg}{AY} \left[\int_0^L dx + \int_0^L \frac{x dx}{L} \right]$$

$$= \frac{Mg}{AY} \left[L + \frac{L^2}{2L} \right]$$

$$= \frac{3MgL}{2AY}$$

18. **Ans (A)**

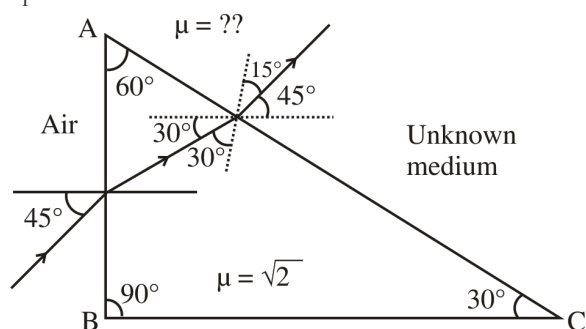
x, y and z have dimensions of velocity.

19. **Ans (B)**

For 1st refraction

$$1 \times \sin 45^\circ = \sqrt{2} \sin r_1$$

$$r_1 = 30^\circ$$



If emergent ray from face AC is parallel to incident ray in air, then using geometry, angle of emergence = 15°

$$\therefore \text{Snell's law : } \sin 30^\circ \times \sqrt{2} = \sin 15^\circ \times \mu$$

$$\therefore \mu = \sqrt{2} \times 2 \cos 15^\circ$$

$$= 2\sqrt{2} \left[\frac{\sqrt{3}+1}{2\sqrt{2}} \right] = \sqrt{3}+1$$

20. **Ans (A)**

Let height of tower be H.

$$-H = 20 \times 6 + \frac{1}{2} (-10) 6^2 \Rightarrow H = 60 \text{ m}$$

$$\therefore \text{Maximum height above ground level} = 60 \text{ m} + 20 \text{ m} = 80 \text{ m}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (2)**

ϕ_2 = Total flux through loop-2

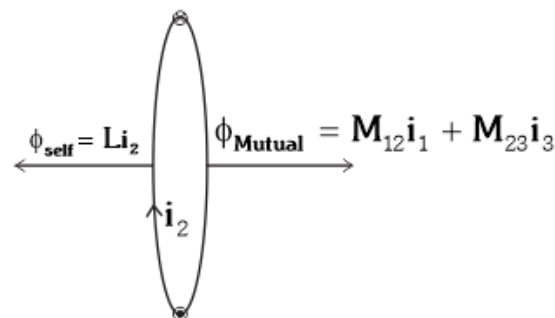
$$\phi_{\text{Self}} + \phi_{\text{Mutual}} =$$

$$= M_{12}i_1 + M_{23}i_3 - Li_2$$

$$\text{EMF induced} = M_{12} \frac{di_1}{dt} + M_{23} \frac{di_3}{dt} - L \frac{di_2}{dt}$$

$$= (2M - L) \frac{di}{dt} = 0 \Rightarrow \frac{L}{M} = 2$$

$$\left[\because \frac{di_1}{dt} = \frac{di_2}{dt} = \frac{di_3}{dt} \text{ and } M_{12} = M_{23} = M \text{ (Symmetry)} \right]$$



2. **Ans (625)**

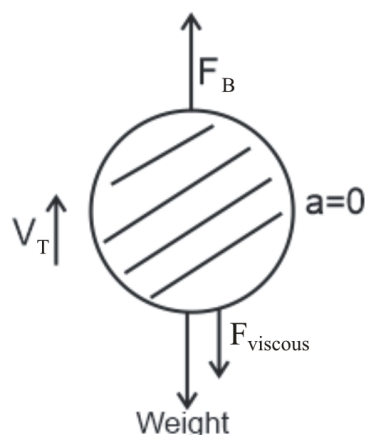
Here terminal speed $v_T = 0.8 \text{ m/sec}$

Balancing forces along vertical direction :

$$F_B = \text{Weight} + F_{\text{viscous}}$$

$$1750 \times \frac{4}{3} \pi R^3 g = 750 \times \frac{4}{3} \pi R^3 g + 6\pi\eta R v_T$$

$$\Rightarrow 1000 \times \frac{4}{3} \pi R^3 g = 6\pi\eta R v_T$$



$$\Rightarrow \eta = \frac{2000R^2g}{9v_T}$$

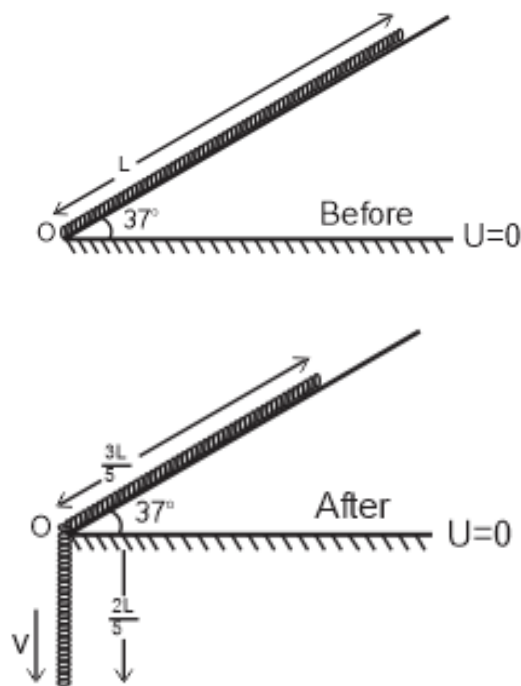
$$\Rightarrow \eta = \frac{2000 \times (1.5 \times 10^{-2})^2 \times 10}{9 \times (0.8)} \text{Decapoise}$$

$$= \frac{5}{8} \text{Decapoise} = \frac{50}{8} \text{Poise}$$

$$\Rightarrow \eta = 6.25 \text{ Poise}$$

$$\therefore 100\eta = 625$$

3. Ans (7)



Neglecting edge effect : T.M.E_{initial} = T.M.E_{Final}

$$Mg \left(\frac{L}{2} \sin 37^\circ \right) + 0 =$$

$$\left(\frac{2M}{5} \right) g \left(\frac{-L}{5} \right) + \left(\frac{3M}{5} \right) g \times \left(\frac{3L}{10} \sin 37^\circ \right)$$

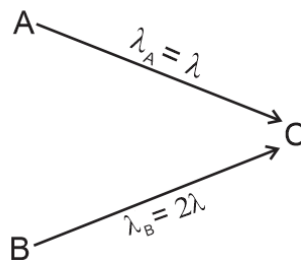
$$+ \frac{1}{2} MV^2$$

$$\Rightarrow \frac{68MgL}{250} = \frac{1}{2} MV^2 \Rightarrow V = \sqrt{\frac{68gL}{125}}$$

$$\Rightarrow V = \sqrt{\frac{68 \times 10 \times 9}{125}} = \sqrt{\frac{68 \times 9 \times 2}{25}} = \frac{\sqrt{1224}}{5}$$

$$\simeq 7 \text{ m/sec (approximated to nearest integer)}$$

4. Ans (3)



Initially \$N_A = N_B = N_0\$

$$\text{So, } \frac{dN_c}{dt} = N_A \lambda_A + N_B \lambda_B$$

$$= N_0 \lambda + N_0 (2\lambda) = 3N_0 \lambda$$

At time \$t = \frac{\ln 2}{\lambda}\$:

$$N_A = N_0 e^{-\lambda \frac{\ln 2}{\lambda}} = \frac{N_0}{2} \text{ and}$$

$$N_B = N_0 e^{-(2\lambda) \times \frac{\ln 2}{\lambda}} = \frac{N_0}{4}$$

\$\therefore\$ Rate of production of C at \$t = \frac{\ln 2}{\lambda}\$ is

$$\frac{dN_c}{dt} = N_A \lambda_A + N_B \lambda_B = \frac{N_0}{2} \lambda + \frac{N_0}{4} 2\lambda = N_0 \lambda$$

\$\therefore x = 3\$

5. Ans (4)

Resultant E. field at any point on the line \$C_1C_2\$ is zero due to all 8 charges.

6. Ans (160)

Temperature, gradient in rod-A = slope of T-x

$$\text{graph} = \frac{4}{3}$$

Temp. gradient in rod-A = slope of T-x graph = 1

Since, the two conductors are in series, heat current through them is equal.

$$K_A A \left(\frac{dT}{dx} \right)_A = K_B A \left(\frac{dT}{dx} \right)_B \Rightarrow 120 \times A \times \frac{4}{3} = K_B \times A \times 1$$

$$\Rightarrow K_B = 160 \text{ J/m-s-}^\circ\text{C}$$

7. Ans (5)

$$E = K^2 e^{-\alpha t}$$

$$dE = 2K e^{-\alpha t} dK + K^2 e^{-\alpha t} (-\alpha dt)$$

$$\frac{dE}{E} = \frac{2dK}{K} + (-\alpha dt)$$

$$\left(\frac{dE}{E} \right)_{\max} \times 100\% = 2 \left(\frac{dK}{K} \right) \times 100\% + \alpha dt \times 100$$

$$= 2 \times 1.75\% + 0.2 \times 1.5t\%$$

$$\left[\frac{dt}{t} \times 100 = 1.5 \Rightarrow dt \times 100 = 1.5t \right]$$

$$= 3.5\% + 0.2 \times 1.5 \times 5\%$$

$$= 5\%$$

8. **Ans (28)**

$$\text{Binding energy} = \Delta mc^2$$

$$\Delta m = m_{\text{expected}} - m_{\text{Actual}} = (2m_p + 2m_n) - (M_{\text{He}} - 2m_e)$$

$$= 2(m_p + m_e) + 2m_n - M_{\text{He}}$$

$$= 2M_{\text{H-atom}} + 2m_n - M_{\text{He}}$$

$$= [2(1.007813) + 2(1.008665) - 4.00260]$$

$$\text{amu}$$

$$= 0.030356 \text{ amu}$$

$$\therefore \text{binding energy} = 0.030356 \times 931 \text{ MeV}$$

$$= 28.26 \text{ MeV}$$

$$= 28 \text{ MeV (To nearest integer)}$$

9. **Ans (40)**

Applying angular momentum conservation about the hinge point, between just before and just after the jerk

$$L_i = L_f$$

$$(m)(\sqrt{2gh})(R) = (m)(\omega R)(R) + \left(\frac{mR^2}{2}\right)\omega$$

$$\omega = \frac{2}{3} \frac{\sqrt{2gh}}{R}$$

$$\omega = \frac{2}{3} \frac{\sqrt{2 \times 10 \times 1.8}}{0.1}$$

$$\omega = 40 \text{ rad/sec}$$

10. **Ans (3)**

The frequency that corresponds to light with 400 nm wavelength is

$$f = c/\lambda = 3 \times 10^8 / 400 \times 10^{-9} = 750 \times 10^{12} \text{ Hz.}$$

The corresponding photon energy is

$$\phi_{\text{ph}} = hf = 6.62 \times 10^{-34} \times 750 \times 10^{12} \text{ J}$$

$$= 497 \times 10^{-21} \text{ J}$$

n = Number of photons incident per second per m^2

$$n = \frac{I}{\phi_{\text{ph}}} = \frac{1000}{497 \times 10^{-21}} = 2.0 \times 10^{21} \text{ photons } \text{m}^{-2} \text{ s}^{-1}.$$

Each photon liberates 1 electron (100% quantum efficiency), thus 2.0×10^{21} electrons $\text{m}^{-2} \text{s}^{-1}$ are circulated. Since the area is 10^{-2} m^2 , the current is $I = 2.0 \times 10^{21} \times 10^{-2} \times 1.6 \times 10^{-19} = 3.22 \text{ A.}$

The short - circuit current delivered by the photocell is 3.22 A.

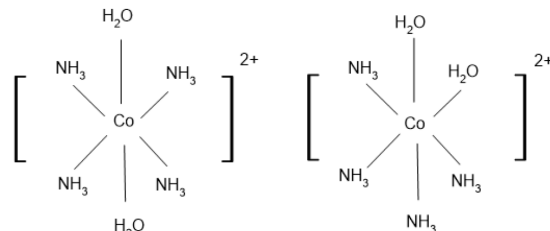
$$i = 3 \text{ A (Approximated to nearest integer).}$$

PART-2 : CHEMISTRY

SECTION-I

1. **Ans (A)**

Incorrect b, c, d



2. **Ans (D)**

Correct - I

Incorrect - II, III, IV

(ii) For hydrogen gas ($P \sim 0, T = 273 \text{ K}$)

At low Pressure, Volume is high $\Rightarrow V \gg b$

For 1 mol,

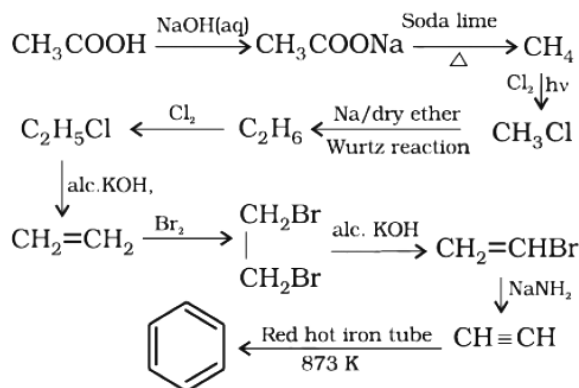
$$\left(P + \frac{a}{V^2}\right)(V - b) = nRT \text{ becomes}$$

$$\left(P + \frac{a}{V^2}\right)V = nRT$$

(iii) Protein has peptide linkages

(iv) Cellulose is a natural polymer

3. **Ans (B)**



4. **Ans (A)**

When concentration of B is doubled keeping the concentration of A fixed, the rate is doubled.

$$r = k [A]^m [B]^1$$

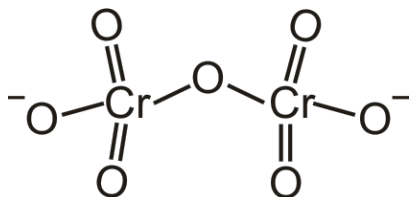
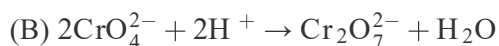
5. **Ans (A)**



In other cases, $\Delta n_g \neq 0$

6. **Ans (C)**

The reaction $\text{MnO}_4^- + \text{NO}_2^- + \text{H}^+ \rightarrow$ gives multiple products. One of the products has trigonal planar structure. This product is NO_3^- .



7. **Ans (B)**

$$\Delta T_f = i K_f m \quad \dots(1)$$

For i,



Bef 1 0

Aft 1- α $\alpha/2$

$$i = \frac{1 - \alpha + (\alpha/2)}{1}$$

Here $\alpha = 0.75$

$$\Rightarrow i = 0.625 \quad \dots(2)$$

From (1), $\Delta T_f = i K_f m$

Where, $m = \frac{75.2/94}{1\text{kg}} \text{ mol}$

$$\Delta T_f = i \times K_f \times m$$

From (2),

$$\Rightarrow 7 = 0.625 \times K_f \times \frac{75.2}{94}$$

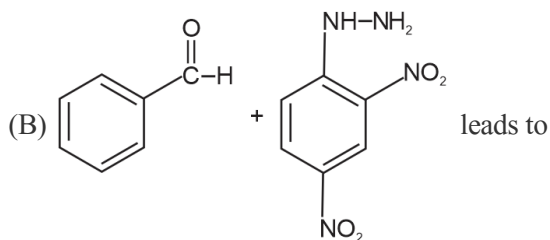
$$\Rightarrow K_f = 14$$

8. **Ans (D)**

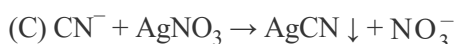
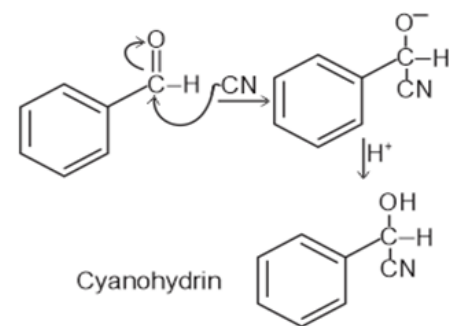
Rate of production of NO_2 is faster when nitric oxide reacts with ozone in the stratosphere.

9. **Ans (C)**

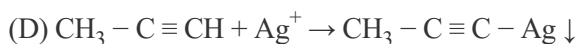
(A) CN^- is a nucleophile



hydrazone derivative rather than cyanohydrin.



white



white

10. **Ans (A)**

Reimer-Tiemann reaction takes place in the presence of the mixture of caustic soda and chloroform.

The electrophile in this reaction is $:\text{CCl}_2$

11. **Ans (B)**

$$\Delta H = \Delta U + \Delta(PV)$$

$$\Rightarrow -560 \text{ kJ} = \Delta U + V \Delta P$$

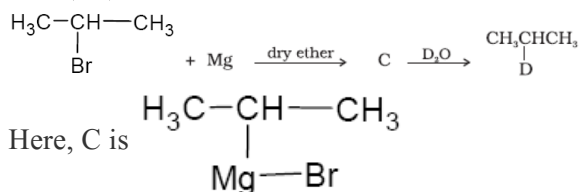
$$\Rightarrow -560 \text{ kJ} = \Delta U + 1\text{L} \times (40 - 70) \text{ atm}$$

$$\Rightarrow -560 \text{ kJ} = \Delta U - 30 \text{ L atm} \times \frac{0.1 \text{ kJ}}{1\text{L atm}}$$

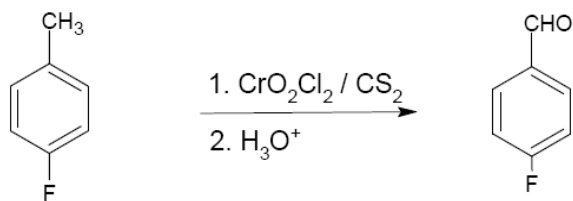
$$\Rightarrow -560 \text{ kJ} = \Delta U - 3 \text{ kJ}$$

$$\Rightarrow -557 \text{ kJ} = \Delta U$$

12. **Ans (B)**



13. Ans (C)



14. Ans (D)

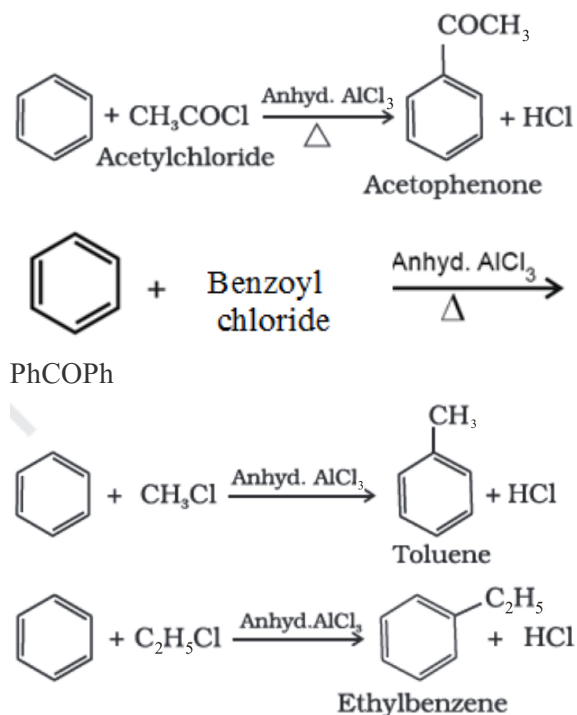
This is structure of Alitame. It is an artificial sweetener.

15. Ans (A)

For the first row transition metals the E values are :

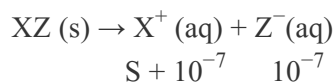
E° (M^{2+}/M)	V	Cr	Mn	Fe	Co	Ni	Cu
	-1.18	-0.91	-1.18	-0.44	-0.28	-0.25	+0.34

16. Ans (C)



17. Ans (A)

$$K_{\text{soln}} = K(X^+) + K(Y^-) + K(Z^-) \quad \dots(1)$$



$$K_{\text{sp}} \\ XY(s) \rightleftharpoons X^+(aq) + Y^-(aq), \\ s + 10^{-7} \quad s$$

At equilibrium,

$$K_{\text{sp}} = Q_{\text{IP}} = [X^+][Y^-] \\ \Rightarrow 12 \times 10^{-14} = (10^{-7} + 7)s \quad \dots(2)$$

Assuming $10^{-7} \gg s$

$$12 \times 10^{-14} = 10^{-7} \times s$$

$$12 \times 10^{-7} = s$$

Hence, assumption invalid

On solving equation (2),

$$s = 3 \times 10^{-7} \Rightarrow [Y^-] = 3 \times 10^{-7} \text{ M}$$

$$[X^+] = 10^{-7} + s = 4 \times 10^{-7} \text{ M}$$

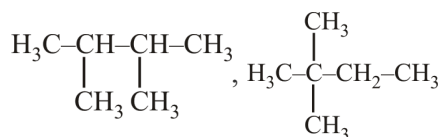
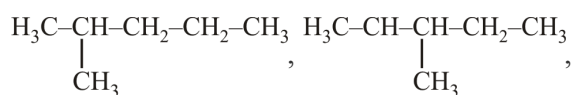
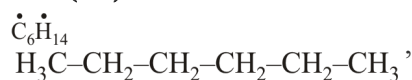
$$\text{for } \Lambda_m^\circ = \frac{K}{[X^+]} \text{ (S.I. units)}$$

$$6 \times 10^{-3} = \frac{K}{4 \times 10^{-7}} \frac{\text{mol}}{\text{L}} \times \frac{1 \text{ L}}{10^3 \text{ m}^3}$$

$$\Rightarrow 24 \times 10^{-10} \text{ S m}^2 \text{ mol}^{-1} = K(X^+) \times 10^{-3} \text{ m}^3 \text{ mol}^{-1}$$

$$\Rightarrow 24 \times 10^{-7} \text{ S m}^{-1} = K(X^+)$$

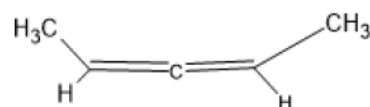
18. Ans (D)



19. Ans (D)

Incorrect (a, b, c)

(a) Following derivative of Allene is chiral without any chiral centre



(b) Blue colour is due to ammoniated electron

(c) Sucrose does not give positive test with Fehling's Solution. It is a non-reducing sugar.

PART-2 : CHEMISTRY

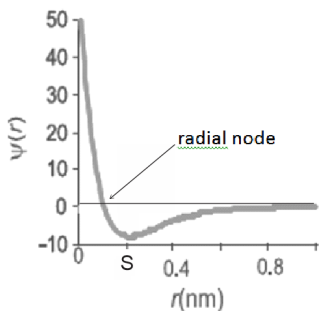
SECTION-II

1. **Ans (1)**

There is 1 radial node

At 0.2 nm, wave function has a negative value.

This orbital wave function of 2s orbital. Hence, Zero angular node.



2. **Ans (3)**

The corrections are as follows :

(i) During the formation of H_2 molecule, a stage is reached where the net force of attraction balances the force of repulsion and system acquires minimum energy.

(ii) Zero overlap

(iii) The actual bond angles of 107° and 104.5° in the NH_3 and H_2O molecules respectively.

Rest are correct.

3. **Ans (801)**

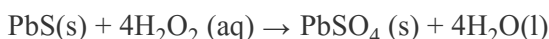
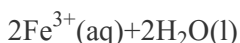
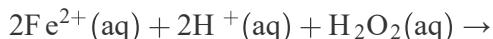
The first ionization enthalpy values (in kJ mol^{-1}) of group 13 elements are :

B Al Ga In Tl

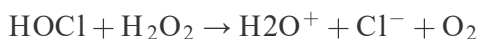
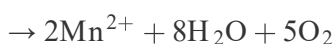
801 577 579 558 589

4. **Ans (2)**

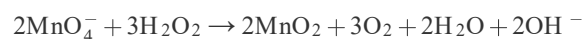
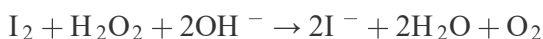
Oxidising action in acidic medium



Reducing action in acidic medium



Reducing action in basic medium



5. **Ans (7)**

H_2CO_3/HCO_3^- buffer system helps to maintain pH of blood between 7.26 to 7.42.

6. **Ans (28)**

Duma's Method is used to estimate nitrogen element.

Gas is N_2

7. **Ans (1)**

Positive	Negative
Hydrated metallic oxides e.g., $Al_2O_3 \cdot xH_2O$, $Cr_2O_3 \cdot xH_2O$ and $Fe_2O_3 \cdot xH_2O$, etc.	Metals e.g., copper, silver gold sols.
Basic dye stuffs, e.g., methylene blue sol.	Metallic sulphides e.g., As_2S_3 , Sb_2S_3 , CdS sols.
Haemoglobin (blood)	Acid dye stuffs e.g., eosin, congo red sols.
oxides, e.g., TiO_2 sol.	Sols of starch, gum, gelatin, clay charcoal, etc.

8. **Ans (2)**

Incorrect - i, ii

Correct- iii, iv, v, vi, vii

Copper is used for making brass (with zinc)

chrome steel for cutting tools & crushing machines

9. **Ans (5)**

Furfural has same number of carbon atoms as those present in D-Xylulose, which is an aketopentose.

10. Ans (1)

Pass H_2S gas through the solution to ensure complete precipitation and separate the precipitate. If the colour of the precipitate is black, it indicates the presence of Cu^{2+} or Pb^{2+} ions. If it is yellow in colour, then presence of As^{3+} ions is indicated.

Group	Cations	Group reagent
Group zero	NH_4^+	None
Group I	Pb^{2+}	Dilute HCl
Group II	Pb^{2+} , Cu^{2+} , As^{3+}	H_2S gas in presence of dil, HCl
Group III	Al^{3+} , Fe^{3+}	NH_4OH in presence of NH_4Cl
Group IV	Co^{2+} , Ni^{2+} , Mn^{2+} , Zn^{2+}	H_2S in presence of NH_4OH
Group V	Ba^{2+} , Sr^{2+} , Ca^{2+}	$(NH_4)_2CO_3$ in presence of NH_4OH
Group VI	Mg^{2+}	None

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

$$n(A \times B) = n(B \times A) = 12000$$

$$n((A \times B) \cap (B \times A)) = (30)^2 = 900$$

$$n((A \times B) \cup (B \times A)) = 12000 + 12000 - 900 = 23100$$

2. Ans (C)

$$\vec{BC} = \frac{\vec{u}}{|\vec{u}|} - \frac{\vec{v}}{|\vec{v}|}, \quad \vec{AC} = \frac{2\vec{u}}{|\vec{u}|}$$

$$\Rightarrow \vec{AB} = \vec{AC} - \vec{BC} = \frac{\vec{u}}{|\vec{u}|} + \frac{\vec{v}}{|\vec{v}|}$$

$$\Rightarrow \vec{AB} \cdot \vec{BC} = \left(\frac{\vec{u}}{|\vec{u}|} + \frac{\vec{v}}{|\vec{v}|} \right) \cdot \left(\frac{\vec{u}}{|\vec{u}|} - \frac{\vec{v}}{|\vec{v}|} \right)$$

$$= \left(\frac{\vec{u} \cdot \vec{u}}{|\vec{u}|^2} - \frac{\vec{v} \cdot \vec{v}}{|\vec{v}|^2} \right)$$

$$\Rightarrow B = \frac{\pi}{2}$$

$$\Rightarrow 1 + \cos 2A + \cos 2B + \cos 2C = 1 - 1 - 4 \cos$$

$$A \cos B \cos C = 0.$$

3. Ans (B)

The given equation is

$$\cos \frac{\pi}{4} (n^2 + 2n) = \sin \frac{\pi}{4} (n^2 + n + 1)$$

$$= \cos \left[\frac{\pi}{2} - \frac{\pi}{4} (n^2 + n + 1) \right] = \cos \frac{\pi}{4} (n^2 + n - 1)$$

$$\Rightarrow \sin \frac{\pi}{8} (2n^2 + 3n - 1) \sin \frac{\pi}{8} (n + 1) = 0$$

$$\text{Either } \sin \frac{\pi}{8} (n + 1) = 0 = \sin k\pi$$

$$\text{i.e. } n + 1 = 8k \Rightarrow n = 8k - 1 \quad k \in I$$

$$\text{or } 2n^2 + 3n - 1 = 8k, k \in I.$$

Let n_1 and n_2 be two integral values of n such that

$$2n_1^2 + 3n_1 - 1 = 8k_1 \quad \dots(1)$$

$$\text{and } 2n_2^2 + 3n_2 - 1 = 8k_2 \quad \dots(2)$$

Subtracting (2) from (1), we get

$$(n_1 - n_2)[2(n_1 + n_2) + 3] = 8(k_1 - k_2).$$

Since $2(n_1 + n_2) + 3$ is odd, $n_1 - n_2$ must be a multiple of 8

$$\text{i.e. } n_1 - n_2 = 8p$$

$$\text{or } n_1 = 8p + n_2.$$

$$\text{For } k_2 = 1, (2) \text{ gives } n_2 = -3. \text{ Hence } n = 8p - 3.$$

Hence the solution is $n = 8p - 1$ or $n = 8p - 3$ for $p \in I$.

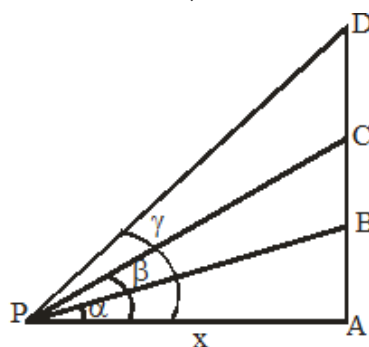
4. Ans (D)

$$AB = x \tan \alpha$$

$$\Rightarrow a = x \tan \alpha$$

$$b = AC = x \tan \beta$$

$$\text{and } c = AD = x \tan \gamma$$



$$\therefore a + b + c = x(\tan \alpha + \tan \beta + \tan \gamma)$$

$$\text{since } \alpha + \beta + \gamma = 180^\circ$$

$$\Rightarrow \tan \alpha + \tan \beta + \tan \gamma = \tan \alpha \cdot \tan \beta \cdot \tan \gamma$$

$$\text{Also } abc = x^3 \tan \alpha \cdot \tan \beta \cdot \tan \gamma$$

$$\Rightarrow x^2 = \frac{abc}{a + b + c}$$

5. **Ans (D)**

$$\text{Let } A \equiv \left(2t_1, \frac{2}{t_1} \right)$$

$$B \equiv \left(2t_2, \frac{2}{t_2} \right)$$

$$C \equiv \left(2t_3, \frac{2}{t_3} \right)$$

$$\text{Slope of } AB = \frac{\frac{2}{t_1} - \frac{2}{t_2}}{2t_1 - 2t_2} = -\frac{1}{t_1 t_2}$$

$$\text{Slope of } AC = -\frac{1}{t_1 t_3}$$

$$\text{Slope of } BC = -\frac{1}{t_2 t_3}$$

$$\angle ACB = 90^\circ \Rightarrow \left(-\frac{1}{t_1 t_3} \right) \times \left(-\frac{1}{t_2 t_3} \right) = -1$$

$$\Rightarrow t_3^2 = \frac{1}{-t_1 t_2}$$

$$\therefore \text{ slope of } AB = t_3^2$$

$$\text{slope of tangent at } C = -\frac{1}{t_3^2}$$

$$\text{As (slope of } AB) \times (\text{slope of tangent at } C) = -1$$

$$\Rightarrow \text{Angle between them} = 90^\circ.$$

6. **Ans (C)**

$$\text{Given expression} = \frac{(\omega^2)^{3n+1} - 1}{\omega^{3n+1} - 1} \cdot \frac{\omega - 1}{\omega^2 - 1}$$

$$= \frac{\omega^2 - 1}{\omega - 1} \cdot \frac{1}{\omega + 1} = 1$$

7. **Ans (B)**

$$(f^2(x))' = 2f(x) \cdot f'(x) \leq 0$$

8. **Ans (D)**

$$\sum_{r=0}^{16} (-1)^r {}^{16}C_r = 2 \sum_{r=0}^6 (-1)^r {}^{16}C_r - 2 \cdot {}^{16}C_7 + {}^{16}C_8 = 0$$

$$\sum_{r=0}^6 (-1)^r {}^{16}C_r = {}^{16}C_7 - \frac{{}^{16}C_8}{2} = {}^{15}C_9$$

$$= 5.7.11.13$$

10. **Ans (D)**

The points (4, 4), (2, 4) and (6, 8), (0, 8) are image of each other in the line $x = 3$.

$\Rightarrow$ The axis of parabola is $x = 3$

$\Rightarrow$ The parabola becomes

$$(x - 3)^2 = 4a(y + \alpha)$$

$$\Rightarrow 1 = 4a(4 + \alpha) \text{ and } 9 = 4a(9 + \alpha)$$

$$\Rightarrow a = \frac{1}{2} \text{ and } \alpha = -7/2$$

$$\Rightarrow (x - 3)^2 = 2 \left(y - \frac{7}{2} \right)$$

is the equation of required parabola.

11. **Ans (A)**

$$\text{We have } A^n = A^{n-2} + A^2 - I$$

$$\Rightarrow A^n - A^{n-2} = A^2 - I.$$

For $n = 49$, this gives

$$A^{49} - A^{47} = A^2 - I.$$

$$\text{Further, } A^{47} - A^{45} = A^2 - I$$

.....

$$A^5 - A^3 = A^2 - I$$

$$A^3 - A = A^2 - I$$

Adding the above equations, we get

$$A^{49} = A + 24 A^2 - 24 I.$$

Here

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

so that

$$A^{49} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 24 & 0 & 0 \\ 24 & 24 & 0 \\ 24 & 0 & 24 \end{bmatrix}$$

$$- \begin{bmatrix} 24 & 0 & 0 \\ 0 & 24 & 0 \\ 0 & 0 & 24 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 25 & 0 & 1 \\ 24 & 1 & 0 \end{bmatrix}$$

12. **Ans (D)**

Radius of given circle $\sqrt{4+2-c} = \sqrt{6-c}$

= a (Say)

Radius of circle $S_1 = \frac{a}{\sqrt{2}}$

Radius of circle $S_2 = \frac{a}{2}$ so on

$$a + \frac{a}{\sqrt{2}} + \frac{a}{2} + \dots \infty = 2$$

$$a \left[\frac{1}{1 - 1/\sqrt{2}} \right] = 2$$

$$a \left(\frac{\sqrt{2}}{\sqrt{2}-1} \right) = 2$$

$$a = 2 - \sqrt{2}$$

$$\Rightarrow \sqrt{6-c} = 2 - \sqrt{2}$$

$$\Rightarrow 6-c = 4 + 2 - 4\sqrt{2}$$

$$\Rightarrow c = 4\sqrt{2}.$$

13. **Ans (B)**

$\phi_1(x)$ and $\phi_2(x)$ both must have a common factor which happens at $k=0$ or $k=30$

$$\Rightarrow p(x) = (x+6)(x-5)(2x+5) \Rightarrow p(0) = -150$$

14. **Ans (D)**

$$I = \int \frac{6(x - x^{-4} + x^{-7}) dx}{(3x^2 + 4 + 2x^{-3} - x^{-6})^2} = \int \frac{dt}{t^2} = \frac{-1}{t} + k$$

$$\text{Where } (3x^2 + 4 + 2x^{-3} - x^{-6}) = t$$

$$\text{So } I = \frac{-x^6}{3x^8 + 4x^6 + 2x^3 - 1} + k$$

15. **Ans (C)**

$$\text{We must have, } f'(x) \geq 0 \forall x \in \mathbb{R} \Rightarrow p^2 \geq \frac{16}{2x^2} \forall$$

$$x \in \mathbb{R}$$

So, $p^2 \geq 16$. (Because maximum value of

$$\frac{16}{2x^2}, x \in \mathbb{R} \text{ is } 16$$

16. **Ans (C)**

We can write $y = A \cos(x+B) - Ce^x$

where $A = c_1 + c_2$, $B = c_3$ and $C = c_4 e^{c_5}$

$$\frac{dy}{dx} = -A \sin(x+B) - Ce^x$$

$$\frac{d^2y}{dx^2} = -A \cos(x+B) - Ce^x$$

$$\Rightarrow \frac{d^2y}{dx^2} + y = -2Ce^x$$

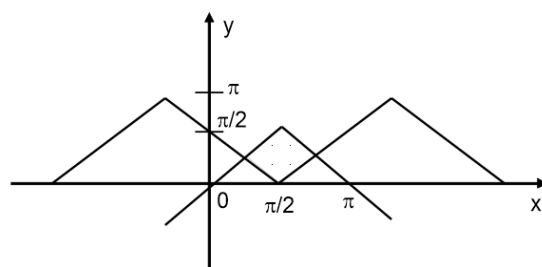
$$\Rightarrow \frac{d^3y}{dx^3} + \frac{dy}{dx} = -2Ce^x = \frac{d^2y}{dx^2} + y$$

$$\Rightarrow \frac{d^3y}{dx^3} - \frac{d^2y}{dx^2} + \frac{dy}{dx} - y = 0$$

Which is a differential equation of degree 1.

17. **Ans (A)**

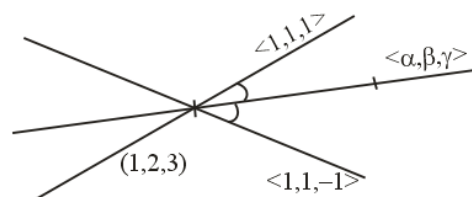
$$\cos^{-1}(\sin x) = \frac{\pi}{2} - \sin^{-1}(\sin x)$$



$$\sin^{-1}(\sin x) = \begin{cases} x, & 0 \leq x < \frac{\pi}{2} \\ \pi - x, & \frac{\pi}{2} \leq x \leq \pi \end{cases}$$

$$\text{Area} = \frac{\pi^2}{8}$$

18. **Ans (A)**



$$\frac{|\alpha + \beta + \gamma|}{\sqrt{3}\sqrt{\alpha^2 + \beta^2 + \gamma^2}} = \frac{\alpha + \beta - \gamma}{\sqrt{3}\sqrt{\alpha^2 + \beta^2 + \gamma^2}}$$

$$\alpha + \beta + \gamma = \alpha + \beta - \gamma \Rightarrow \gamma = 0$$

$$\text{OR } \alpha + \beta + \gamma = -\alpha - \beta + \gamma \Rightarrow \alpha + \beta = 0$$

So option A is correct

19. Ans (A)

$$2f(x) + f(-x) = \frac{1}{x} \sin\left(\frac{x^2-1}{x}\right) \quad \dots(1)$$

By replacing x to $-x$ we get

$$2f(-x) + f(x) = \frac{1}{x} \sin\left(\frac{x^2-1}{x}\right) \quad \dots(2)$$

So by equation (1) and (2) we get

$$f(-x) = f(x) = \frac{1}{3x} \sin\left(\frac{x^2-1}{x}\right)$$

$$\text{So } I = \int_{\frac{1}{e}}^e f(x) dx = \int_{\frac{1}{e}}^e \frac{1}{3x} \sin\left(\frac{x^2-1}{x}\right) dx$$

Now put $x = \frac{1}{t}$ we get

$$I = \int_1^{\frac{1}{e}} \frac{t}{3} \sin\left(\frac{1}{t} - t\right) \left(-\frac{1}{t^2}\right) dt = -I$$

$$\Rightarrow 2I = 0 \text{ or } I = 0$$

20. Ans (A)

$$t_n = \left[\left(\frac{n+1}{n} \right)^{n+1} - \frac{n+1}{n} \right]^{-n}$$

$$\therefore (t_n)^{1/n} = \left[\left(\frac{n+1}{n} \right)^{n+1} - \frac{n+1}{n} \right]^{-1}$$

$$= \left(\frac{n+1}{n} \right)^{-1} \left[\left(1 + \frac{1}{n} \right)^n - 1 \right]^{-1}$$

$$\therefore \lim_{n \rightarrow \infty} (t_n)^{1/n} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^{-1} \left[\left(1 + \frac{1}{n} \right)^n - 1 \right]^{-1}$$

$$= 1 \times (e - 1)^{-1}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1)

$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

Substituting values of $\cos A$ and $\sin A$ from the given equations, we get

$$= \frac{\sin B \cos^2 B \cos B - (\cos B + \cos^2 B) \sin B}{\sqrt{2}} = \frac{-\sin B \cos B}{\sqrt{2}}$$

Squaring and adding given equations

$$(\cos B + \cos^3 B)^2 + (\sin B - \sin^3 B)^2 = 2$$

$$\Rightarrow \cos^2 B + \cos^6 B + 2 \cos^4 B +$$

$$\sin^2 B + \sin^6 B - 2 \sin^4 B = 2$$

$$\Rightarrow 1 + 2(\cos^2 B + \sin^2 B) +$$

$$(\sin^4 B + \cos^4 B - \sin^2 B \cos^2 B) = 2$$

$$\Rightarrow 1 + 2 \cos^2 B - 2 \sin^2 B + 1 - 3 \sin^2 B \cos^2 B = 2$$

$$\Rightarrow 3 \cos^2 2B + 8 \cos 2B - 3 = 0$$

$$\Rightarrow \cos 2B = \frac{1}{3}$$

$$\Rightarrow \sin 2B = \pm \frac{2\sqrt{2}}{3}$$

$$\Rightarrow \sin(A - B) = -\frac{\sin 2B}{2\sqrt{2}} = \pm \frac{1}{3}$$

2. Ans (1)

Using :

$$L = \left| \lim_{x \rightarrow \frac{\pi}{4}} \frac{4(2 \sin x \cos^2 x - \sin^3 x) + \sin x + \cos x}{\cos x - \sin x} \right|$$

$$= |-1| = 1$$

3. Ans (6)

Family of lines can be written as

$$(3x + 7y + 11) \sec \theta + \operatorname{cosec} \theta (5x - 3y - 11) = 0$$

$$\Rightarrow (5x - 3y - 11) + \tan \theta (3x + 7y + 11) = 0$$

So, for any value of θ this line passes through the point of intersection of lines

$$5x - 3y - 11 = 0 \text{ and } 3x + 7y + 11 = 0 \text{ i.e. from P } (1, -2).$$

Let $A(x, y)$ be a point on the line $x - y + 3 = 0$

Applying cosines rule in ΔPAB

$$PB^2 = PA^2 + AB^2 - 2PA \cdot AB \cos \theta$$

$$\geq PA^2 + AB^2 - 2PA \cdot AB = (PA - AB)^2$$

$$\Rightarrow (PA - AB)^2 \leq PB^2$$

So, maximum value of $|PA - PB|$ is

$$PB = \sqrt{4 + 36} = 2\sqrt{10}.$$

4. **Ans (2)**

Let E_1 – A wins the match at the end of 11th game

E_2 – B win exactly one game.

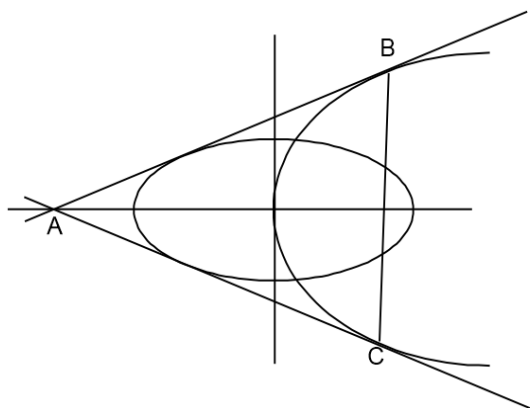
A can win the match at the end of 11th game, by winning exactly one of the first 10 games and draws the remaining the remaining 9 or by winning exactly one of the first 10 games, loses exactly one and draws remaining 8.

$$P(E_1) = {}^{10}C_1 \left(\frac{1}{2}\right) \left(\frac{1}{3}\right)^9 \times \frac{1}{2} + {}^{10}C_2 \left(\frac{1}{2}\right) \left(\frac{1}{3}\right)^8 \left(\frac{1}{6}\right) (2!) \left(\frac{1}{2}\right)$$

$$\therefore P\left(\frac{E_2}{E_1}\right) = \frac{{}^{10}C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{3}\right)^8 \left(\frac{1}{6}\right) (2!)}{{}^{10}C_1 \left(\frac{1}{2}\right) \left(\frac{1}{3}\right)^9 + {}^{10}C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{3}\right)^8 \left(\frac{1}{6}\right) (2!)}$$

$$= \frac{9}{11}$$

5. **Ans (5)**



Let the common tangents be AB and AC. The common tangents would intersect at x-axis due to symmetry

$\Rightarrow$ Triangle ABC is equilateral if $\angle BAC = 60^\circ$

$\Rightarrow m_{AB} = \tan 30^\circ = \frac{1}{\sqrt{3}}$ where m_{AB} is the slope

of the line AB.

Since line AB is tangent to both the ellipse and the parabola.

$$\Rightarrow \frac{1}{m} = \sqrt{4m^2 + b^2}$$

$$\Rightarrow 3 = \frac{4}{3} + b^2$$

$$\Rightarrow b^2 = \frac{5}{3} \Rightarrow b = \sqrt{\frac{5}{3}}$$

6. **Ans (288)**

Sum of the digits at even places can be 6, 9, 12 or 15. Corresponding to each way in table, digits at even places can be arranged in 3! ways and remaining three digits can be arranged in 3! ways. So total number of such numbers = $8 \times 3! \times 3! = 288$

Sum	Digits
6	1,2,3
9	1,2,6 1,3,5 2,3,4
12	1,5,6 2,4,6 3,4,5
15	4,5,6
	Total = 8 ways

7. **Ans (8)**

Point z will be point of intersection of diagonals of quadrilateral formed by points $(0, 0)$, $(1, 0)$, $(1, 1)$ & $(0, 1)$

8. **Ans (7)**

$$\lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n \frac{1}{(2r-1)(2r+1)(2r+3)(2r+5)}}{\sum_{r=1}^n \frac{1}{(4r-1)(4r+3)(4r+7)(4r+11)}}$$

$$= \lim_{n \rightarrow \infty} \frac{\frac{1}{6} \left[\frac{1}{1 \cdot 3 \cdot 5} - \frac{1}{(2n+1)(2n+3)(2n+5)} \right]}{\frac{1}{12} \left[\frac{1}{3 \cdot 7 \cdot 11} - \frac{1}{(4n+3)(4n+7)(4n+11)} \right]}$$

$$= \frac{12 \cdot 3 \cdot 7 \cdot 11}{6 \cdot 1 \cdot 3 \cdot 5}$$

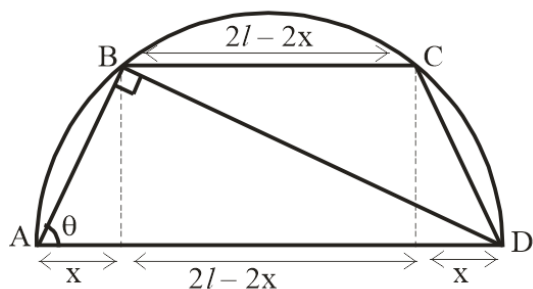
$$\text{So, } k = \frac{154}{5}$$

$$\text{Now, } \frac{5}{22} k = \frac{5}{22} \cdot \frac{154}{5} = 7$$

9. Ans (60)

$$AB = 2l \cos \theta \text{ from } \triangle ABD$$

$$\therefore x = AB \cos \theta = 2l \cos^2 \theta$$



$$\text{Perimeter} = 2l + 2l - 2x + 2AB$$

$$= 4l - 2x + 4l \cos \theta$$

$$P(\theta) = 4l(1 - \cos^2 \theta + \cos \theta)$$

$$\therefore P'(\theta) = 4l(2 \cos \theta \sin \theta - \sin \theta) = 0$$

$$\therefore 2 \cos \theta \sin \theta = \sin \theta \quad \theta \neq 0 \text{ or } \pi$$

$$\therefore \cos \theta = \frac{1}{2}$$

$$\therefore \theta = \pi/3 = 60^\circ$$

$$\therefore \text{Perimeter is maximum when } \theta = 60^\circ \text{ Ans.}]$$

10. Ans (5)

$$\text{Let } (x_i - 8) = d_i$$

$$\therefore \sigma_x = \sigma_d = \sqrt{\frac{\sum d_i^2}{n} - \left(\frac{\sum d_i}{n}\right)^2} = \sqrt{\frac{45}{18} - \left(\frac{9}{18}\right)^2} = \frac{3}{2}$$